Which Physics-Informed Formulation?

A controlled screening study on the heterogeneous faulted tidal aquifer benchmark

Generated	2026-08-08 01:51
Reference	MODFLOW 6, 100 x 100 cells, 48 stress periods of 2 h
Runs	42 completed
Budget	240 s wall clock per run, 4 CPU threads
Design	one factor at a time from a common baseline
Axes	form, arch, fault, temporal, balance, kfield

floor	mass_local_err_frac	0.07875
floor	mass_global_err_frac	0.02119
floor	mass_global_err_m3d	1.752e+05
floor	river_total_rel_err	6.025e-07
floor	river_cell_rmse_m3d	0.01309
floor	river_nse	1
floor	river_sign_agreement	1
null	null_head_rmse_m	0.9125
null	null_fault_jump_rmse_m	0.5584

Results

HOW TO READ THESE TABLES

Every metric is "lower is better". Each run got the same wall-clock budget, so the iteration count is a result, not a control: a formulation that is cheap per step gets more steps, and that is part of its merit. The floors listed on the title page are what the MODFLOW reference itself scores under the same metrics - a surrogate reaching the floor is perfect as far as this study can measure.

HEAD SIMULATION (HEAD RMSE (M))

name	score_head	skill_vs_null	iters	head_nse	fault_all_jump_recovery	k_pattern_corr
combo:fv+sidefeat	0.7027	0.2299	1743	0.994	0.9244	NaN
form:fv	0.7455	0.183	1804	0.9933	0.9332	NaN
fv+causal-norm:eps2	0.7543	0.1734	1742	0.9931	0.9337	NaN
form:fv	0.7724	0.1535	1794	0.9928	0.9335	NaN
combo:fv+pirate	0.775	0.1507	1070	0.9927	0.9299	NaN
combo:fv+causal	0.7934	0.1305	1333	0.9924	0.9343	NaN
form:fv	0.8133	0.1088	1790	0.992	0.9252	NaN
budget:fv	0.8264	0.09434	7015	0.9917	0.9279	NaN
-- persistence (null) --	0.9125	0	NaN	0.9899	NaN	NaN

HEAD ACROSS FAULTS (RMSE OF THE HEAD JUMP (M))

name	score_fault	skill_vs_null	iters	head_nse	fault_all_jump_recovery	k_pattern_corr
form:fv	0.557	0.002464	1794	0.9928	0.9335	NaN
combo:fv+causal	0.557	0.002389	1333	0.9924	0.9343	NaN
form:fv	0.5571	0.002217	1804	0.9933	0.9332	NaN
budget:fv	0.5583	0.0001821	7015	0.9917	0.9279	NaN
-- persistence (null) --	0.5584	0	NaN	NaN	0.9277	NaN
arch:mlp+rwf	0.5594	-0.001917	4034	0.9893	0.928	NaN
fv+causal-norm:eps2	0.5602	-0.003192	1742	0.9931	0.9337	NaN
form:mixed	0.5608	-0.004323	4150	0.9711	0.9361	NaN
form:mixed	0.5611	-0.004897	3344	0.972	0.9338	NaN

MASS BALANCE (LOCAL IMBALANCE FRACTION)

Results (continued)

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name	score_mass	iters	head_nse	fault_all_jump_recovery	k_pattern_corr
budget:fv	0.5028	7015	0.9917	0.9279	NaN
form:fv	0.6341	1790	0.992	0.9252	NaN
fv+causal-norm:eps2	0.6663	1742	0.9931	0.9337	NaN
form:fv	0.6723	1794	0.9928	0.9335	NaN
form:fv	0.679	1804	0.9933	0.9332	NaN
combo:fv+causal	0.6942	1333	0.9924	0.9343	NaN
combo:fv+sidefeat	0.7002	1743	0.994	0.9244	NaN
combo:fv+pirate	0.7271	1070	0.9927	0.9299	NaN

RIVER EXCHANGE (RELATIVE ERROR IN TOTAL EXCHANGE)

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name	score_river	iters	head_nse	fault_all_jump_recovery	k_pattern_corr
mixed+sidefeat:rescaled	0.1203	4085	0.9688	0.9304	NaN
baseline	0.1307	2049	0.979	0.9179	NaN
budget:strong	0.1327	8419	0.9807	0.926	NaN
causal-norm:eps1	0.1519	2073	0.9786	0.9344	NaN
arch:mlp+rwf	0.1609	4034	0.9893	0.928	NaN
bal:fixed	0.1612	2197	0.9781	0.934	NaN
form:mixed	0.1641	4150	0.9711	0.9361	NaN
mixed:rescaled	0.1707	4348	0.9684	0.9387	NaN

INVERSE (K RECOVERY) (LOG10 RMSE OF K)

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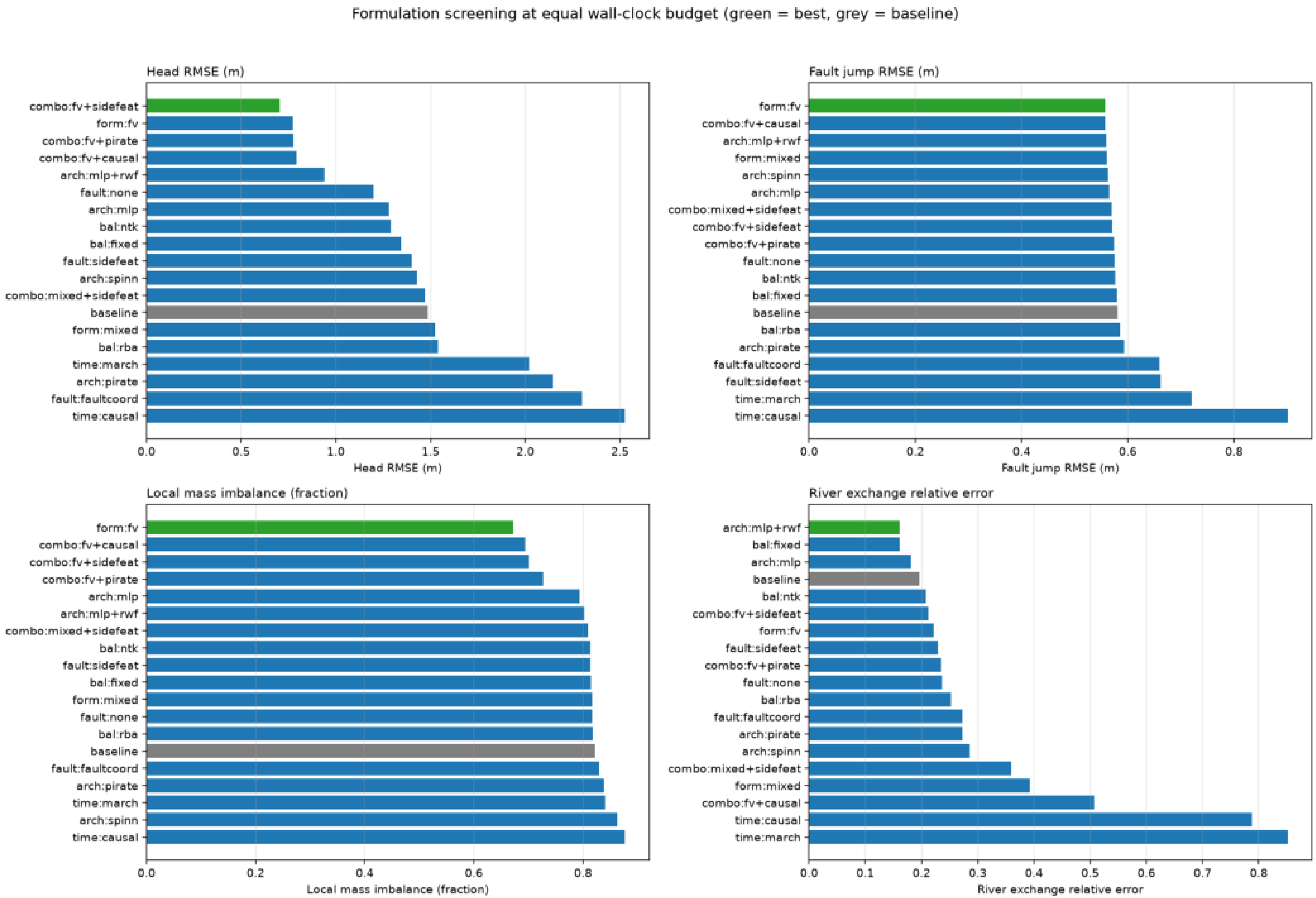
name	score_inverse	iters	head_nse	fault_all_jump_recovery	k_pattern_corr
inv:fv+kl	0.8534	1114	0.9945	0.9362	0.1055
inv:grid	0.8633	2110	0.9895	0.9542	0.04202
inv:fv+grid	0.8691	1726	0.9954	0.9367	0.2175
inv:kl	0.9751	1820	0.9904	0.9509	0.1022
inv:net	2.647	1900	0.9883	0.9548	0.04562

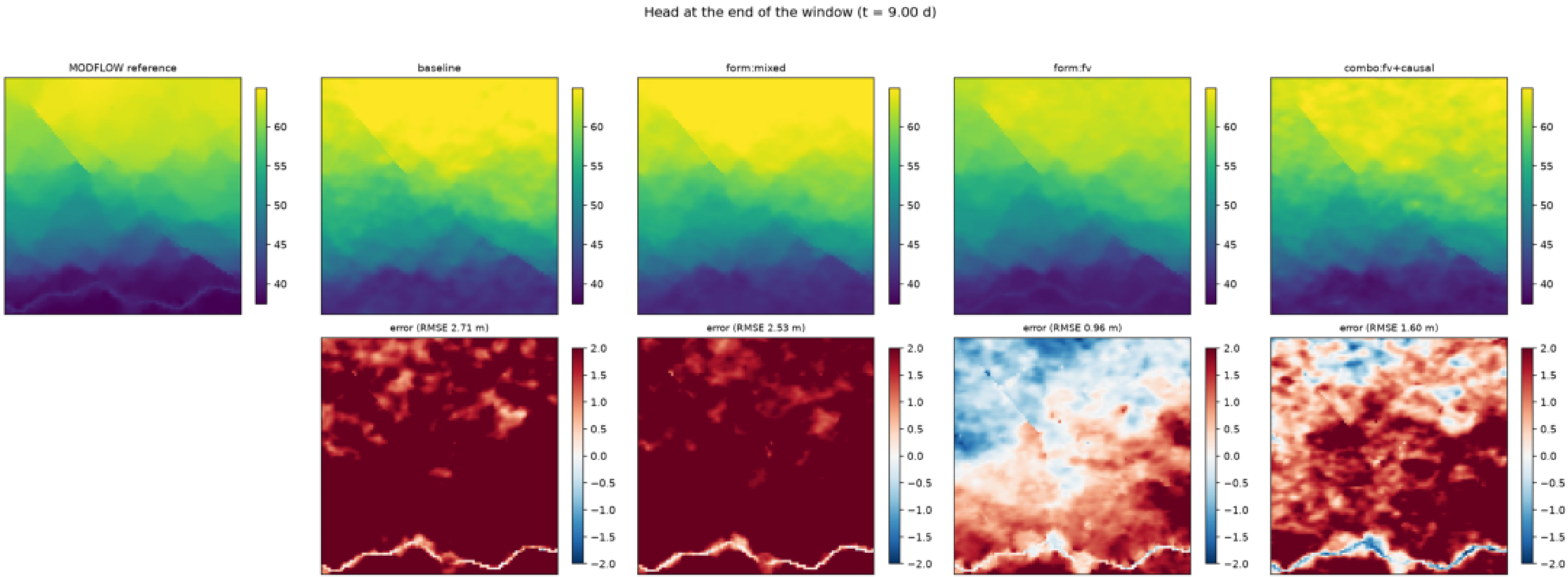
EFFECT OF EACH AXIS VS THE BASELINE (% CHANGE, NEGATIVE IS BETTER)

Results (continued)

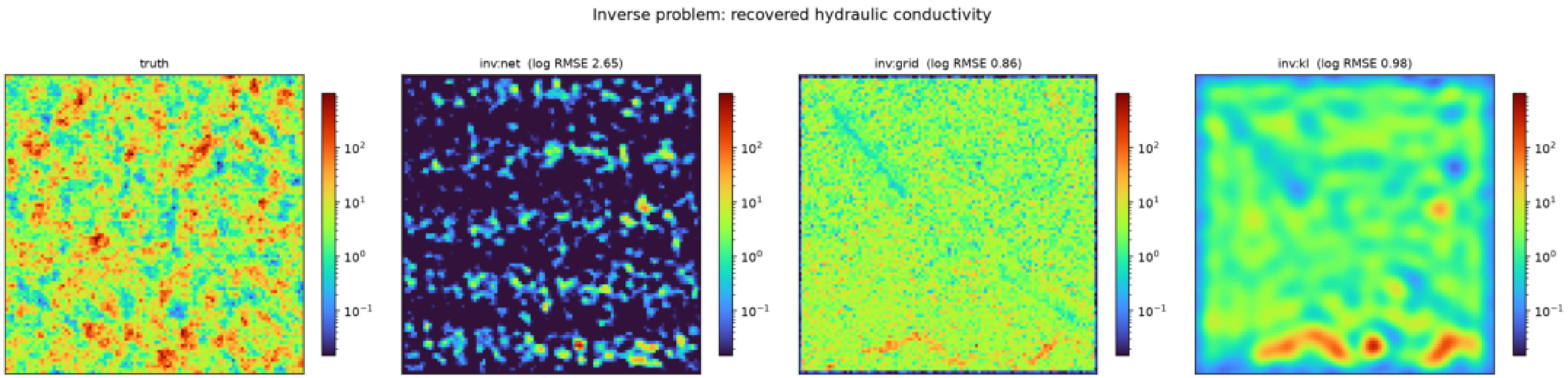
name	group	head	fault	mass	river
form:mixed	form	2.4	-3.3	-0.7	99.6
form:fv	form	-48.0	-4.0	-18.2	12.8
arch:mlp	arch	-13.9	-2.6	-3.4	-7.5
arch:pirate	arch	44.3	2.2	2.0	38.7
arch:spinn	arch	-3.8	-3.1	4.9	45.0
arch:mlp+rwf	arch	-36.7	-3.6	-2.5	-18.1
fault:none	fault	-19.3	-0.8	-0.6	20.3
fault:sidefeat	fault	-5.8	13.9	-1.1	16.7
fault:faultcoord	fault	54.8	13.7	0.9	38.6
time:causal	temporal	70.1	55.3	6.7	300.8
time:march	temporal	36.1	24.0	2.2	333.1
bal:fixed	balance	-9.5	-0.0	-1.0	-18.0
bal:ntk	balance	-13.1	-0.6	-1.1	5.7
bal:rba	balance	3.5	0.8	-0.5	28.3
combo:fv+causal	combo	-46.6	-4.0	-15.6	158.3
combo:fv+sidefeat	combo	-52.7	-1.6	-14.8	8.1
combo:mixed+sidefeat	combo	-1.1	-1.9	-1.6	82.9
combo:fv+pirate	combo	-47.8	-1.1	-11.6	19.2
mixed:rescaled	followup	8.8	-0.8	-3.5	-13.2
mixed+sidefeat:rescaled	followup	8.1	3.0	-3.2	-38.8
causal:eps0.1	followup	86.8	25.2	5.1	445.2
causal:eps0.01	followup	36.5	0.5	1.4	27.6
budget:strong	budget	-15.1	-1.4	-7.1	-32.5
budget:fv	budget	-44.4	-3.8	-38.9	4.1
causal-norm:eps1	causal	-10.5	0.0	-1.1	-22.7
causal-norm:eps5	causal	11.5	-0.6	0.1	-4.3
fv+causal-norm:eps2	causal	-49.2	-3.5	-19.0	8.1
form:mixed	form	9.1	-0.9	-3.7	-11.0
form:fv	form	-49.8	-4.0	-17.4	18.0
form:mixed	form	26.0	-3.0	0.3	29.4
form:fv	form	-34.5	-3.1	-3.6	55.5
form:mixed	form	4.1	-3.4	-3.8	-16.6
form:fv	form	-45.3	-3.0	-22.9	27.6



Screening result per criterion; green = best, grey = baseline.



Head field at the end of the window, and the error against MODFLOW.



Recovered hydraulic conductivity for each parameterisation.